

CONNER GRIFFIN

connergriffin.github.io ◦ cgrffn11 at memphis dot edu ◦ 601-543-6964

Research Statement

1. INTRODUCTION

My primary research interests are in Ramsey theory and combinatorial number theory. To date, I have completed five research projects which have resulted in one publication [10] and one paper in preparation [11]. In Ramsey theory, there are “density” and “partition” theorems. This refers to the given conditions in the associated theorems. Consider two cornerstone results of the field.

Theorem 1.1 (van der Waerden’s Theorem [18]). *Let $\mathbb{N} = \bigcup_{i=1}^n A_i$ be a finite partition of $\mathbb{N} = \{1, 2, \dots\}$. Then there exists an $1 \leq i \leq n$ such that for all $k \in \mathbb{N}$ there are a and b in $\mathbb{N}$ such that*

$$\{a, a + b, a + 2b, \dots, a + (k - 1)b\} \subset A_i.$$

Definition 1.2. For $A \subseteq \mathbb{N}$, define **the upper density of A** to be

$$\bar{d}(A) := \limsup_N \frac{|A \cap [N]|}{N},$$

where $[N] = \{1, 2, \dots, N\}$.

Theorem 1.3 (Szemerédi’s Theorem [17]). *Let $A \subseteq \mathbb{N}$ be such that $\bar{d}(A) > 0$. Then for all $k \in \mathbb{N}$ there are a and b in $\mathbb{N}$ such that*

$$\{a, a + b, a + 2b, \dots, a + (k - 1)b\} \subset A.$$

Van der Waerden’s theorem is an example of a partition result and Szemerédi’s theorem is its density analogue. It is clear that $\bar{d}(A) > 0$ is saying something about the size of A in $\mathbb{N}$. As it turns out, van der Waerden’s Theorem and other partition theorems can be framed in terms of large subsets of $\mathbb{N}$. The typical argument is that for any partition of $\mathbb{N}$ some cell of the partition is *large* enough – which can be interpreted using an appropriate algebraic notion of size – to guarantee the sought algebraic configuration. This was first done in the case of van der Waerden’s theorem by Kakeya and Morimoto. [15]

As a researcher, my goal is to solve problems in a way that is complete and provides insight into the nature of the problem. To this end, I study the history, background, and development of the problems I am working on with the intention of developing a comprehensive perspective. I believe this quality is apparent in the work outlined here. All of which deals with modern approaches to generalizations of van der Waerden’s and Szemerédi’s Theorems.

2. RESEARCH PROJECTS

2.1. Filter Relative Notions of Size and Combinatorial Consequences. In four of the five research projects I have completed, I performed a careful analysis of new, broad notions of size. These ideas have a very strong connection with the Stone-Čech compactification of a semigroup, and all of the study being done is through the clarifying perspective it provides.

2.1.1. Introduction to the Stone-Čech compactification and filter relative notions of size. Filter relative notions of size are generalizations of the following properties – each relevant in interconnected ways to Ramsey theory.

Definition 2.1. Let $A \subseteq \mathbb{N}$. For $k \in \mathbb{N}$, take $-k + A := \{x \in \mathbb{N} : x + k \in A\}$. We say that

- (1) A is **syndetic** if it has bounded gaps. More technically, $\exists k \in \mathbb{N}, \bigcup_{i=1}^k -k + A = \mathbb{N}$.
- (2) A is **thick** if it contains arbitrarily long sequences of consecutive numbers. More technically, $\forall k \in \mathbb{N}, \bigcap_{i=1}^k -k + A \neq \emptyset$.
- (3) A is **piecewise syndetic** if there is some bound k such that A contains arbitrarily long sequences of numbers with gaps between successive numbers no longer than k . More technically,

$$\exists k \in \mathbb{N}, \left\{ -x + \bigcup_{i=1}^k (-i + A) : x \in \mathbb{N} \right\}$$

has the finite intersection property.

The “more technical” parts of the above definitions have been appropriately generalized to arbitrary semigroups. See [13] for more on this. Important to the study of Ramsey theory, piecewise syndetic sets form a *prime* collection of sets. That is if $A \cup B$ is piecewise syndetic then either A is piecewise syndetic or B is piecewise syndetic.

The *filter relative* generalization of syndeticity, thickness and piecewise syndeticity allow us to localize their combinatorial structure to a particular filter – whose sets might have strong arithmetic properties themselves! Research directions involving these filtered notions were initiated by Shuungula, Zelenyuk and Zelenyuk in [16].

Definition 2.2. Let $\mathcal{F}$ be a filter on a semigroup S – whose operation we will denote by juxtaposition – and $A \subseteq S$. For all $x \in S$ and $A \subseteq S$, define $x^{-1}A := \{y \in S : xy \in A\}$. We say that,

- (1) A is **$\mathcal{F}$ -syndetic** if $\forall V \in \mathcal{F}, \exists H \in \mathcal{P}_f(V), \bigcup_{h \in H} h^{-1}A \in \mathcal{F}$.
- (2) A is **$\mathcal{F}$ -thick** if $\exists V \in \mathcal{F}, \forall H \in \mathcal{P}_f(V), \{\bigcap_{h \in H} h^{-1}A\} \cup \mathcal{F}$ has the finite intersection property.
- (3) A is **piecewise $\mathcal{F}$ -syndetic** if

$$\forall V \in \mathcal{F}, \exists H \in \mathcal{P}_f(V), U_V \in \mathcal{F}, \left\{ x^{-1} \bigcup_{h \in H} h^{-1}A : x \in U_V, V \in \mathcal{F} \right\} \cup \mathcal{F}$$

has the finite intersection property.

The Stone-Ćech compactification of a discrete topological space, X , is homeomorphic to the space of all ultrafilters on X with the correct topology. It is denoted βX . For an arbitrary semigroup S , the ultrafilter product extends the semigroup operation on S to βS .

Definition 2.3. Let p, q be ultrafilters on a semigroup S . Then define

$$p \cdot q := \{A \subseteq S : \{x \in S : x^{-1}A \in q\} \in p\}.$$

For many mathematicians first introduced to this product, it can be helpful to point out that from a meaningful perspective this product corresponds to convolution of measures. A set $A \subseteq S$ is piecewise syndetic if and only if it belongs to a $p \in K(\beta S)$, the unique minimal ideal of βS . There is a fourth class of large sets, **central sets**, which are exactly the sets belonging to *idempotents* in $K(\beta S)$. Likewise, **$\mathcal{F}$ -central sets** belong to *idempotents* in $K(\overline{\mathcal{F}})$. Often, any results that can be found using the Stone-Ćech compactification of a semigroup can readily be extended to a filter relative version in which we assume that $\overline{\mathcal{F}}$ is a subsemigroup. This is because much of the same theoretical background that the βS results rely on are true for a closed subsemigroup $\overline{\mathcal{F}} := \{p \in \beta S : \mathcal{F} \subseteq p\}$.

2.1.2. An infinitary nilpotent polynomial Hales-Jewett theorem. The first project I undertook for my dissertation was an infinitary nilpotent polynomial Hales-Jewett theorem. In [14], Johnson and Richter

prove a nilpotent polynomial Hales-Jewett theorem making extensive use of piecewise $\mathcal{F}$ -syndetic sets. Of course, $\mathcal{F}$ -central sets are piecewise $\mathcal{F}$ -syndetic and thus must contain whatever configurations piecewise $\mathcal{F}$ -syndetic sets contain. An infinitary generalization of their result follows from properties of $\mathcal{F}$ -central sets.

Definition 2.4. We call $\mathcal{R} \subseteq \mathcal{P}_f(\mathbb{N})$ an *IP-ring* if there is a sequence, $(\alpha_i)_{i=1}^\infty$, with $\alpha_i < \alpha_j$ for all $i < j$ such that

$$\mathcal{R} = \text{FU} \left((\alpha_i)_{i=1}^\mathbb{N} \right) := \left\{ \bigcup_{i \in H} \alpha_i : H \in \mathcal{P}_f(\mathbb{N}) \right\}.$$

Theorem 2.5 (Infinitary nilpotent polynomial Hales-Jewett (Griffin, 2023)). *Let $\mathcal{F}$ be an idempotent filter on a nilpotent group Γ and let $\mathbb{P}$ be a good collection of $\mathcal{F}$ -measurable polynomial mappings. For all finite choices of $g_1, \dots, g_k \in \mathbb{P}$, all $\mathcal{F}$ -central sets E there exists an IP-ring $\mathcal{R}$ and an IP-set $(v_\alpha)_{\alpha \in \mathcal{R}} \subseteq \Gamma$ with*

$$\{g_1(\alpha)v_\alpha, \dots, g_k(\alpha)v_\alpha\}_{\alpha \in \mathcal{R}} \subseteq E.$$

2.1.3. A combinatorial characterization of $\mathcal{F}$ -central sets. In [8], Goswami and Poddar prove an $\mathcal{F}$ -central sets theorem. A natural next step is a combinatorial characterization of $\mathcal{F}$ -central sets which I took on as my second project. This is a filter relative generalization of the work done by Hindman, Maleki, and Strauss in [12].

An FP-tree, T , in $A \subseteq S$ is a tree in $\Lambda := \cup_{n=1}^\infty S^n$ such that the image of all $f \in T$ is in A and for all $f \in T$, $B_f := \{x : f \frown x \in T\} = \cup_{f \frown g \in T} \text{FP}(g)$ where $\frown$ represents concatenation of sequences and $\text{FP}(\cdot)$ is the set of all finite products over a sequence.

Definition 2.6. Let S be a semigroup. Then $\mathcal{A} \subset \mathcal{P}(S)$ is **collectionwise piecewise $\mathcal{F}$ -syndetic** if there is a $p \in K(\overline{\mathcal{F}})$ such that $\mathcal{A} \subseteq p$.

Theorem 2.7 (Combinatorial Characterization of $\mathcal{F}$ -Central Sets (Griffin, 2023)). *Let S be a semigroup, let $\mathcal{F}$ be a filter on S such that $\overline{\mathcal{F}}$ is a subsemigroup of βS , and let $A \subseteq S$. Then A is $\mathcal{F}$ -central if and only if there is an FP-tree, T , in A such that $\{B_f : f \in T\}$ is collectionwise piecewise $\mathcal{F}$ -syndetic.*

Theorem 2.7 was independently achieved in [1]. Leveraging the full strength of [13, Theorem 4.20], I also prove the following, which is new in even the non-filter-relative case established in [12].

Definition 2.8 (Simple product tree). Let S be a semigroup and T a tree in $A \subseteq S$. Then T is a **simple product tree** if and only if $f \in T$ implies $\text{FP}(f) \subseteq A$.

Theorem 2.9 (Simplified Combinatorial Characterization (Griffin, 2023)). *Let S be a semigroup, let $\mathcal{F}$ be a filter on S such that $\overline{\mathcal{F}}$ is a subsemigroup of βS , and let $A \subseteq S$. Then A is $\mathcal{F}$ -central if and only if there is a simple product tree, T , in A such that $\{B_f : f \in T\}$ is collectionwise piecewise $\mathcal{F}$ -syndetic.*

2.1.4. A characterization of piecewise $\mathcal{F}$ -syndetic sets. The following theorem is well known and incredibly useful. The known proofs used only elementary set theory which, as far as is currently known, cannot be adapted to the filter relative case. This meant that proving a filter relative generalization required an original argument.

Theorem 2.10. *Let S be a semigroup. Then $A \subseteq S$ is piecewise syndetic if and only if there is a syndetic sets B and a thick set C such that $A = B \cap C$.*

In [5], Christopherson and Johnson prove that the intersection of an $\mathcal{F}$ -syndetic set and an $\mathcal{F}$ -thick set is piecewise $\mathcal{F}$ -syndetic. They left the converse as an open question which I took on as my third project.

In order to achieve a solution to this problem, I developed a new perspective on relative notions of size. We are taking $\text{Syn}(\mathcal{F})$, $\text{Thick}(\mathcal{F})$, and $\text{PS}(\mathcal{F})$ to be the collection of all $\mathcal{F}$ -syndetic, $\mathcal{F}$ -thick, and piecewise $\mathcal{F}$ -syndetic subsets of S , respectively. We define, for all $\mathcal{F} \subseteq \mathcal{P}_f(S)$, $\mathcal{F}^* := \{A \subseteq S : A^c \notin \mathcal{F}\}$.

Theorem 2.11. *Let S be a set, $\mathcal{F}$ a filter on S . Then*

$$\bigcup_{p \in \overline{\mathcal{F}}} p = \mathcal{F}^* \text{ and } \bigcap_{p \in \overline{\mathcal{F}}} p = \mathcal{F}$$

Theorem 2.12 (Griffin, 2023). *Let $\mathcal{F}$ be a filter on a semigroup S . Then*

$$\text{Syn}(\mathcal{F}) = \bigcap_{q \in \overline{\mathcal{F}}} \bigcup_{p \in \overline{\mathcal{F}}} (p \cdot q) \text{ and } \text{Thick}(\mathcal{F}) = \bigcup_{q \in \overline{\mathcal{F}}} \bigcap_{p \in \overline{\mathcal{F}}} (p \cdot q).$$

Theorem 2.13 (Griffin, 2023). *Let $\mathcal{F}$ be a filter on a semigroup S . Then*

$$\text{PS}(\mathcal{F}) = \bigcup_{r \in \overline{\mathcal{F}}} \bigcap_{q \in \overline{\mathcal{F}}} \bigcup_{p \in \overline{\mathcal{F}}} (p \cdot q \cdot r)$$

The completion of the filter relative version of Theorem 2.10 followed from a strengthening of Theorem 2.13 in which r may be taken to be a minimal idempotent. This amounts to a strengthening of the critically important [13, Theorem 4.43].

Theorem 2.14 (Griffin, 2023). *Let $\mathcal{F}$ be a filter on S such that $\overline{\mathcal{F}}$ is a subsemigroup of βS . If A is piecewise $\mathcal{F}$ -syndetic, then A is the intersection of an $\mathcal{F}$ -syndetic set and an $\mathcal{F}$ -thick set.*

2.1.5. Subsemigroup actions and $(\mathcal{F}, \mathcal{G})$ -central sets. This is joint work with Shea Burns, Dennis Davenport, Shakuan Frankson, John H. Johnson, and Malik Kebe. It was supported by the American Institute for Mathematics' SQuaRE program. It is currently in preparation. All of the following have been defined previously, with the exception of piecewise $(\mathcal{F}, \mathcal{G})$ -syndeticity. Such a notion has been defined in other work but in a way that is inequivalent.

Definition 2.15. Let $\mathcal{F}, \mathcal{G}$ be a filter on a semigroup S and $A \subseteq S$. We say that

- (1) A is **$(\mathcal{F}, \mathcal{G})$ -syndetic** if $\forall V \in \mathcal{F}, \exists H \in \mathcal{P}_f(V), \bigcup_{h \in H} h^{-1}A \in \mathcal{G}$.
- (2) A is **$(\mathcal{F}, \mathcal{G})$ -thick** if $\exists V \in \mathcal{F}, \forall H \in \mathcal{P}_f(V), \{\bigcap_{h \in H} h^{-1}A\} \cup \mathcal{G}$ has the finite intersection property.
- (3) A is **piecewise $(\mathcal{F}, \mathcal{G})$ -syndetic** if

$$\forall V \in \mathcal{F}, \exists H \in \mathcal{P}_f(V), U_V \in \mathcal{G}, \{x^{-1} \bigcup_{h \in H} h^{-1}A : x \in U_V, V \in \mathcal{F}\} \cup \mathcal{G}$$

has the finite intersection property.

We define $K(\mathcal{F}, \mathcal{G}) = \cup\{\overline{\mathcal{F}} \cdot e : e \in E(K(\overline{\mathcal{G}}))\}$, and $\text{PS}(\mathcal{F}, \mathcal{G})$ to be the set of all piecewise $(\mathcal{F}, \mathcal{G})$ -syndetic sets.

Assuming $\overline{\mathcal{G}}$ is a subsemigroup and $\overline{\mathcal{F}} \cdot \overline{\mathcal{G}} \subseteq \overline{\mathcal{F}}$, $K(\mathcal{F}, \mathcal{G})$ may be shown to contain idempotents.

Definition 2.16. A set $A \subseteq S$ is **$(\mathcal{F}, \mathcal{G})$ -central** if there is an idempotent $e \in K(\mathcal{F}, \mathcal{G})$ such that $A \in e$.

We have shown that $(\mathcal{F}, \mathcal{G})$ -central sets satisfy a strong central sets theorem and can be combinatorially characterized.

2.2. Density Ramsey Theorems. For the fourth project in my dissertation, I prove a density Ramsey theorem. As stated here, it is a partial result towards a two dimensional Szemerédi theorem for a finitely generated free semigroup. The result has been proved in arbitrary finite dimensions. The main tool in this case is ergodic theory. In particular, I made use of conditional expectation, disintegration of measure, and the ergodic Szemerédi theorem for commuting transformations of Furstenberg and Katznelson. [6]

Definition 2.17. Let Λ be a finite set. The **free semigroup generated by Λ** is $\Lambda^* = \bigcup_{n=0}^{\infty} \Lambda^n$ with the semigroup operation being concatenation.

Definition 2.18. Let $S_r(\Lambda^*) := \{w \in \Lambda^* : |w| = r\}$. Then define, for $A \subseteq \Lambda^*$, the upper density of A to be

$$\bar{d}(A) = \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{r=0}^{N-1} \frac{|A \cap S_r(\Lambda^*)|}{|S_r(\Lambda^*)|}$$

Definition 2.19. We say that $w' \in \Lambda^*$ is a descendant of w if there exists a $u \in \Lambda^*$ such that $w' = wu$. An **arithmetic subtree** in $A \subseteq \Lambda^*$ of order r is a map $\phi : B_r(\Lambda^*) \rightarrow A$ such that there exists $q \in \mathbb{Z}_{>0}$ such that for all $w \in B_{r-1}(\Lambda^*)$ and $\lambda \in \Lambda$ we have

- (1) $|\phi(w\lambda)| = |\phi(w)| + q$
- (2) The element $\phi(w\lambda)$ is a descendent of $\phi(w)\lambda$.

We call q the gap of this arithmetic subtree.

Theorem 2.20 (Szemerédi's theorem for arithmetic trees (Furstenberg and Weiss,[7])). *If $A \subseteq \Lambda^*$ is such that $\bar{d}(A) > 0$, then for all $r \in \mathbb{N}$ there is an arithmetic tree of order r in A .*

A significant partial result toward finding the product of two arbitrarily long arithmetic trees with the same gap in positive density subsets of $\Gamma := \Lambda^* \oplus \Lambda^*$ was achieved by Bulinski and Fish in [4].

The **free product** $\tilde{\Gamma} = \Lambda^* * \Lambda^*$ is the free semigroup generated by $\{X_\lambda\}_{\lambda \in \Lambda} \sqcup \{Y_\lambda\}_{\lambda \in \Lambda}$. Suppose $\gamma \in \tilde{\Gamma}$ has n X -letters and m Y -letters, then the **level of γ** is $L(\gamma) := (n, m)$.

Definition 2.21. Given $A \subseteq \Gamma$ and $u, v \in \mathbb{Z}_{\geq 0}^2$, we define a (u, v) -arithmetic product tree of order r in A to be a map $\phi : B_r(\tilde{\Gamma}) \rightarrow A$ such that:

- (1) For $\gamma \in B_{r-1}(\tilde{\Gamma})$ and $\lambda \in \Lambda$ we have

$$L(\phi(\gamma X_\lambda)) = L(\phi(\gamma)) + u$$

and

$$L(\phi(\gamma Y_\lambda)) = L(\phi(\gamma)) + v$$

- (2) For $\gamma \in B_{r-1}(\tilde{\Gamma})$ and $\lambda \in \Lambda$ we have that $\phi(\gamma X_\lambda)$ is a descendent of $\phi(\gamma)X_\lambda$ and $\phi(\gamma Y_\lambda)$ is a descendent of $\phi(\gamma)Y_\lambda$

Theorem 2.22. [4, Theorem 1.7] *Let $A \subseteq \Gamma$ be a subset with $\bar{d}(A) > 0$ and let $u, v \in \mathbb{Z}_{>0}^2$. Then for $r > 0$ there exists $n \in \mathbb{Z}_{>0}$ such that there is an (nu, nv) -arithmetic product tree of order r in A .*

2.2.1. A two dimensional corner theorem and more for arithmetic trees. We can prove the following in arbitrary finite dimensions. For simplicity, we state a two dimensional version. A manuscript of this work is currently in preparation.

Theorem 2.23 (Griffin, 2024). *Let $A \subseteq \Gamma$ be a subset with $\bar{d}(A) > 0$. Then for all $r > 0$ there is an n and a $x \in A$ such that, in A , there is an $((n, 0), (0, 0))$ and a $((0, 0), (0, n))$ arithmetic product tree each*

with order r rooted at x . Moreover, for each vertex, y , in these trees, for any $u, v \in \mathbb{Z}_{>0}^2$, and any $r_y > 0$, there is an (nu, nv) arithmetic product tree of order r_y in A rooted at y .

In the result from Bulinski and Fish, they take $u, v \in \mathbb{Z}_{>0}^2$ which means that as one moves from one generation to the next in the product trees one must move “diagonally” in Γ . For example, a corners theorem like the one below cannot be deduced directly from their result. The generation after (w, x) in the first coordinate would look like (wiy_i, xa_i) where a_i has length equal to n times the second position of u . We partly single out this much smaller structure because of the role a “simplices” theorem – of a stronger nature – plays in combinatorial proofs of the multidimensional Szemerédi theorem. [9, Theorem 10.3]

Corollary 2.24 (Griffin, 2024). *Suppose $A \subseteq \Gamma$ is such that $\bar{d}(A) > 0$. Then there exist $w, x \in \Lambda^*$ and $n \in \mathbb{N}$ such that for all $i \in \Lambda$ there exists $y_i, z_i \in \Lambda^n$ such that $\{(w, x), (wiy_i, x), (w, xiz_i)\}_{i \in \Lambda} \subseteq A$.*

3. CURRENT PROJECTS AND FUTURE WORK

3.1. Filter Relative Notions of Size and Combinatorial Consequences.

3.1.1. Multiple structures and relative notions. This is joint work with Sayan Goswami and Sourav Patra. We are developing some combinatorial results involving sums and products for partitions of the interval $(0, 1)$ whose cells are either all Lebesgue measurable or all have the property of Baire. This is the setting laid out by Bergelson, Hindman, and Weiss in [3] and then further developed by Bergelson, Hindman and Leader in [2]. It seems there are several results that we will be able to prove here.

3.1.2. A hierarchy of notions of size in a semigroup. In the analytic and arithmetic hierarchies, the number of times one alternates quantifiers in a formula the higher the complexity of the sets definable by those formulas. This is reflected in the new perspective I introduced of syndeticity, thickness and piecewise syndeticity. Inspired by this we define the following.

Definition 3.1. Let $n \in \mathbb{Z}_{\geq 0}$ and S be a semigroup. Let $\mathcal{F}$ be a filter on S . We define

$$\sigma^n = \{A \subseteq S : (\exists p_{2n+1} \in \overline{\mathcal{F}}), (\forall p_{2n} \in \overline{\mathcal{F}}), \dots, (\exists p_1 \in \overline{\mathcal{F}}) \ A \in p_1 \cdot p_2 \cdot \dots \cdot p_{2n+1}\}$$

and

$$\tau^n = \{A \subseteq S : (\forall p_{2n+1} \in \overline{\mathcal{F}}), (\exists p_{2n} \in \overline{\mathcal{F}}), \dots, (\forall p_1 \in \overline{\mathcal{F}}) \ A \in p_1 \cdot p_2 \cdot \dots \cdot p_{2n+1}\}$$

Notice that $\sigma^0 = \mathcal{F}^*$, $\tau^0 = \mathcal{F}$, $\sigma^1 = \text{PS}(\mathcal{F})$ and that for all n , $(\sigma^n)^* = \tau^n$. It is also the case that τ^n is a filter and σ^n is prime. That is if $A \cup B \in \sigma^n$ then either $A \in \sigma^n$ or $B \in \sigma^n$.

We can define σ^n combinatorially. From this definition and other things, some interesting partial results follow – both when $\overline{\mathcal{F}}$ is a semigroup and when it is *not* a semigroup.

Proposition 3.2 (Griffin, 2024). *When $\overline{\mathcal{F}}$ is a semigroup, for all $n \geq 1$, $\sigma^0 \supsetneq \sigma^1 = \sigma^n$ and $\tau^0 \subsetneq \tau^1 = \tau^n$.*

3.2. Density Ramsey Theorems.

3.2.1. A multidimensional Szemerédi theorem for arithmetic trees. One would hope to answer the following question either positively or negatively.

Question 3.3. For any $A \subseteq \Gamma := \Lambda^* \oplus \Lambda^*$ with positive upper density and any $r \in \mathbb{N}$ are there two arithmetic trees of order r , ϕ and ψ , each having the same gap n such that $\phi(B_r(\Lambda^*)) \times \psi(B_r(\Lambda^*)) \subseteq A$?

This is a challenging question. As Bulinski and Fish point out in [4], if one can find an $((n, 0), (0, n))$ -arithmetic product tree which factors with the natural homomorphism from $\tilde{\Gamma}$ to Γ , then Question 3.3 would be answered positively.

REFERENCES

- [1] L. L. Baglini, S. K. Patra, and M. M. Shaikh. Dynamical notions along filters. *New York Journal of Mathematics*, 29:792–817, 2023.
- [2] V. Bergelson, N. Hindman, and I. Leader. Additive and multiplicative Ramsey theory in the reals and the rationals. *Journal of Combinatorial Theory, Series A*, 85(1):41–68, 1999.
- [3] V. Bergelson, N. Hindman, and B. Weiss. All-sum sets in $(0, 1]$ – Category and measure. *Mathematika*, 44(1):61–87, 1997.
- [4] K. Bulinski and A. Fish. Arithmetic subtrees in large subsets of products of trees. *Journal D’Analyse Mathématique*, 150(3):359–381, 2023.
- [5] C. Christopherson and J. H. Johnson. Algebraic characterizations of some relative notions of size. *Semigroup Forum*, 104:28–44, 2022.
- [6] H. Furstenberg and Y. Katznelson. An ergodic Szemerédi theorem for commuting transformations. *Journal D’Analyse Mathématique*, 34(1):275–291, 1978.
- [7] H. Furstenberg and B. Weiss. Markov processes and Ramsey theory for trees. *Combinatorics, Probability and Computing*, 12(5–6):547–563, 2003.
- [8] S. Goswami and J. Poddar. Central sets theorem along filters and some combinatorial consequences. *Indagationes Mathematicae*, 33(6):1312–1325, 2022.
- [9] W. T. Gowers. Hypergraph regularity and the multidimensional Szemerédi theorem. *Annals of Mathematics*, 166:897–946, 2007.
- [10] C. Griffin. A characterization of piecewise $\mathcal{F}$ -syndetic sets. *Semigroup Forum*, 109(2):375–389, 2024.
- [11] C. Griffin and et. al. Subsemigroup actions of βS and $(\mathcal{F}, \mathcal{G})$ -central sets (working title). In preparation.
- [12] N. Hindman, A. Maleki, and D. Strauss. Central sets and their combinatorial characterization. *Journal of Combinatorial Theory, Series A*, 74(2):188–208, 1996.
- [13] N. Hindman and D. Strauss. *Algebra in the Stone-Čech Compactification*. volume 27 of De Gruyter Expositions in Mathematics. Walter de Gruyter, Berlin – New York, 2012.
- [14] J. H. Johnson and F. K. Richter. Revisiting the nilpotent polynomial Hales–Jewett theorem. *Advances in Mathematics*, 321:269–286, 2017.
- [15] S. Kakeya and S. Morimoto. On a theorem of MM. Bandet and van der Waerden. *Japanese journal of mathematics*, 7:163–165, 1930.
- [16] O. Shuungula, Y. Zelenyuk, and Y. Zelenyuk. The closure of the smallest ideal of an ultrafilter semigroup. *Semigroup Forum*, 79(3):531–539, 2009.
- [17] E. Szemerédi. On sets of integers containing no k elements in arithmetic progression. *Acta Arithmetica*, 27:199–245, 1975.
- [18] B. L. van der Waerden. Beweis einer baudetschen vermutung. *Nieuw. Arch. Wisk.*, 15, 1927.